A Project Report on

# CivicMind – AI Based Centralized Municipal Grievance Management System

submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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(2025-2026)

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# CERTIFICATE

This is to certify that the project entitled **"CivicMind – AI Based Centralized Municipal Grievance Management System"** is a bonafide work of **Umesh Phulare (24206002), Pranay Bhere (24206004), Vinay Patil (23106072), Ved Sawant (24206001)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**)

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# Project Report Approval for T. E.

This project report entitled ***CivicMind – AI Based Centralized Municipal Grievance Management System*** by ***Umesh Phulare, Pranay Bhere, Vinay Patil, Ved Sawant*** is approved for the degree of ***Bachelor of Engineering*** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning)**, ***2025-26***.

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# Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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# Abstract

With the rapid growth of urban populations, efficient grievance management has become a critical requirement for modern smart cities. Traditional complaint handling systems are often slow, manual, and lack transparency, resulting in delayed responses and reduced citizen satisfaction. To address these challenges, this project presents CivicMind, an AI-powered centralized grievance management system designed to automate and enhance complaint handling. The system integrates advanced technologies such as Natural Language Processing (NLP) and Computer Vision (CNN) to enable intelligent classification, urgency detection, and image-based validation of complaints. Citizens can submit complaints along with images, textual descriptions, and location data through a user-friendly interface. The system performs validation to ensure consistency between inputs, reducing misleading reports, and includes a deduplication mechanism using text similarity and geospatial analysis. CivicMind incorporates intelligent routing to assign complaints to appropriate departments based on workload and availability, improving efficiency. It also provides real-time tracking, notifications, and feedback mechanisms to enhance transparency and citizen engagement. Experimental results show high accuracy and reduced response time. Overall, the system offers a scalable and efficient solution for urban governance and aligns with Sustainable Development Goals by supporting innovation in governance (SDG 9), promoting efficient urban environments (SDG 11), and enhancing transparency and accountability (SDG 16).

**Keywords:** Artificial Intelligence, Smart City, Grievance Management System, Natural Language Processing (NLP), Computer Vision (CNN), Complaint Classification, Image Validation, Deduplication, Intelligent Routing, Real-Time Tracking

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## ABBREVIATIONS

AI ....................................................*Artificial Intelligence*
API.................................................. *Application Programming Interface*
CNN................................................ *Convolutional Neural Network*
DB................................................... *Database*
GPS................................................. *Global Positioning System*
HTTP ............................................. *Hypertext Transfer Protocol*
I/O .................................................. *Input/Output*
JSON .............................................. *JavaScript Object Notation*
ML ................................................. *Machine Learning*
JWT ................................................ *JSON Web Token*
SDG ............................................... *Sustainable Development Goal*
UI ................................................... *User Interface*
NLP................................................. *Natural Language Processing*
REST .............................................. *Representational State Transfer*

# Chapter 1

# Introduction

## 1.1 Background

Rapid urbanization has significantly increased the complexity of managing city-level infrastructure and public services, especially in developing countries like India. With urban populations rising rapidly, municipal corporations are facing a surge in citizen complaints related to roads, sanitation, water supply, electricity, waste management, and public safety. According to recent estimates, India’s urban population is expected to reach nearly 600 million by 2031, placing immense pressure on civic administration systems.

Traditional grievance redressal systems are largely manual and inefficient. These systems typically rely on paper-based submissions, manual categorization, and human-driven assignment of complaints. As a result, they suffer from several major issues such as delayed response times, duplicate complaints, lack of transparency, and poor resource utilization. Citizens often remain unaware of the status of their complaints, leading to dissatisfaction and reduced trust in governance systems.

Existing digital platforms have attempted to address some of these challenges by enabling online complaint submission and tracking. However, most of these systems function merely as digitized versions of manual workflows without incorporating intelligent automation.

They lack capabilities such as automated classification, fraud detection, urgency prioritization, and workload balancing, which are essential for modern smart city governance.

With the advancement of Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), and Computer Vision, there is an opportunity to transform traditional governance systems into intelligent, automated, and citizen-centric platforms. These technologies can enable automated complaint processing, real-time validation, and efficient resource allocation.

The proposed system, CivicMind, is an AI-enabled centralized municipal grievance management system designed to address these challenges. It integrates advanced AI techniques to automate the entire complaint lifecycle—from submission to resolution—while improving transparency, efficiency, and citizen engagement.

## 1.2 Motivation

The motivation behind developing CivicMind arises from the critical limitations observed in existing grievance management systems and the growing need for smarter governance solutions. Current systems suffer from multiple inefficiencies, including:

- Lack of image authenticity verification, allowing fake or irrelevant complaints
- Absence of urgency detection, resulting in equal prioritization of all complaints
- Manual complaint assignment, creating operational bottlenecks
- No workload balancing among municipal employees
- Limited transparency and poor communication with citizens

These issues not only delay problem resolution but also impact public safety, especially in critical cases such as electrical hazards, water leakage, or road damage. Additionally, inefficient complaint handling leads to resource wastage and reduces the overall effectiveness of municipal operations.

At the same time, modern AI technologies have demonstrated strong capabilities in text classification, image recognition, and decision automation. By leveraging these advancements, it is possible to design a system that can intelligently process complaints, detect fraud, prioritize urgent cases, and allocate resources efficiently. Furthermore, there is a growing demand for citizen-centric governance platforms that promote transparency and engagement. A system that allows real-time tracking, automated updates, and clear communication can significantly improve public trust and participation in governance.

Thus, the motivation of this project is to develop an intelligent, scalable, and automated grievance management system that bridges the gap between citizens and municipal authorities while enhancing operational efficiency and governance transparency.

## 1.3 Aim & Scope of the Project

The primary aim of this project is to design and develop an AI-based centralized municipal grievance management system, namely CivicMind, that enhances the efficiency, transparency, and responsiveness of urban governance. The system aims to automate the complete lifecycle of complaint handling, including submission, verification, classification, routing, and tracking. By integrating advanced technologies such as Natural Language Processing (NLP) and Convolutional Neural Networks (CNN), the platform provides intelligent decision-making capabilities while reducing dependency on manual administrative processes.

The scope of the project includes the development of a web-based platform that facilitates seamless interaction between citizens and municipal authorities. The system enables users to submit complaints using textual descriptions, images, and real-time location data. It supports automated classification of complaints into relevant sectors and urgency levels, ensuring that critical issues receive timely attention. Furthermore, the platform incorporates image validation mechanisms to verify the authenticity and relevance of submitted complaints, thereby reducing fraudulent or misleading reports.

In addition, the system provides intelligent routing and assignment of complaints to appropriate departments and available personnel based on workload, expertise, and geographical proximity. The platform also ensures real-time tracking and status updates, allowing citizens to monitor the progress of their complaints, thereby improving transparency and accountability in governance.

The project primarily focuses on developing a scalable and efficient solution suitable for deployment across multiple municipal corporations. It is designed to handle moderate to large volumes of complaints and can be extended for integration with smart city infrastructure. The system is particularly relevant for urban local bodies aiming to adopt digital governance solutions and improve citizen engagement through technology-driven services.

## 1.4 Relevance to Sustainable Development Goals

The proposed CivicMind system contributes to the advancement of digital governance and aligns with several United Nations Sustainable Development Goals (SDGs). The project strongly supports SDG 9: Industry, Innovation and Infrastructure by promoting the development of intelligent digital infrastructure for urban governance. By integrating AI-driven automation into municipal systems, CivicMind enhances operational efficiency and supports the modernization of public service delivery.

Additionally, the system aligns with SDG 11: Sustainable Cities and Communities, as it aims to improve urban living conditions by ensuring faster resolution of civic issues such as sanitation, road maintenance, and public safety. Efficient complaint management contributes to cleaner, safer, and more resilient cities.

Furthermore, the project supports SDG 16: Peace, Justice and Strong Institutions by improving transparency, accountability, and citizen engagement in governance. Real-time tracking, automated workflows, and data-driven decision-making help build trust between citizens and public institutions.

# Chapter 2

# Literature Survey

The rapid growth of urban populations, coupled with increasing demands for efficient public service delivery, has significantly intensified the need for robust grievance management systems in modern smart cities. As cities expand, citizens encounter a wide range of civic issues such as road damage, waste management problems, water leakage, and infrastructure failures. Traditional complaint handling mechanisms, which rely heavily on manual processes such as physical visits to municipal offices, phone calls, or basic web portals, are often inefficient, time-consuming, and lack proper coordination. These systems frequently suffer from issues such as delayed responses, misclassification of complaints, lack of transparency in processing, and poor communication between citizens and authorities, ultimately leading to reduced citizen satisfaction and trust in governance.

To address these challenges, there has been a growing shift towards the adoption of digital and intelligent grievance management systems. These systems aim to streamline the entire lifecycle of complaint handling, including submission, classification, prioritization, routing, and resolution. In recent years, advancements in Artificial Intelligence (AI), particularly in Natural Language Processing (NLP) and Computer Vision, have enabled the automation of key processes within these systems. NLP techniques allow systems to analyze and interpret textual complaint data, identify relevant categories, and determine urgency levels, while Computer Vision techniques enable the analysis of images submitted by users to verify and understand the nature of civic issues.

The integration of these technologies has significantly improved the efficiency, accuracy, and scalability of grievance management systems. Automated systems reduce dependency on manual intervention, minimize human error, and enable faster decision-making. Moreover, they enhance transparency by providing real-time updates and tracking mechanisms for citizens. This chapter presents a comprehensive review of existing research works and systems in the domain of civic grievance management, focusing on their methodologies, technological approaches, strengths, and limitations. The analysis provides a foundation for identifying research gaps and motivates the need for an advanced, integrated solution such as CivicMind.

## 2.1 Smart City Complaint Management Systems

Early systems primarily focused on providing digital platforms for complaint submission and tracking. Pooja et al. [5] proposed a smart city complaint management system using a web and Android application that allows citizens to submit complaints along with images, text descriptions, and location data. The system improves communication between citizens and municipal authorities and reduces the need for physical visits. However, it relies heavily on manual processing and lacks intelligent automation.

Similarly, the CivicWatch platform proposed by Kishore et al. [6] introduces a community-driven approach where users can report issues with geotagged images and track their resolution. The system enhances transparency and citizen engagement through real-time tracking and validation mechanisms. Despite these improvements, the system still depends on manual categorization and lacks advanced AI-based decision-making capabilities.

## 2.2 AI-Based Complaint Classification and Routing

Recent research has focused on integrating Artificial Intelligence to automate complaint processing. Shastri et al. [1] proposed an AI-powered complaint analyzer using a Distil-BERT framework, which performs both classification and urgency detection. The system significantly reduces response time and improves routing accuracy through automated decision-making.

Similarly, Saha et al. [2] developed a system that uses Natural Language Processing and machine learning techniques such as TF-IDF and BERT for complaint classification and routing. Their approach reduces manual workload and improves efficiency in handling large volumes of complaints.

Ashwitha and Vani [3] introduced a smart civic complaint analyzer that uses NLP techniques to classify complaints, detect urgency levels, and provide analytical insights through dashboards. The system demonstrates improved transparency and faster processing compared to traditional systems. However, these systems primarily focus on textual data and do not incorporate image-based validation.

## 2.3 Image-Based Civic Issue Detection

To address the limitations of text-based systems, several studies have explored image-based analysis. Nimase et al. [7] proposed a system that uses image processing and Convolutional Neural Networks (CNN) to classify civic issues and route them to appropriate departments. This approach reduces manual effort in identifying the nature of complaints.

A more advanced approach is presented by Kumar et al. [4], where scene graph models are used to understand civic issues from images. The system generates structured representations of objects and relationships within images, enabling better interpretation of complex civic problems. This research highlights the importance of integrating visual data for accurate complaint analysis.

## 2.4 Complaint Deduplication and Similarity Analysis

Handling duplicate complaints is a significant challenge in large-scale systems. Cui et al. [8] proposed a similarity-based approach using case-based reasoning to retrieve similar complaint cases and improve resolution efficiency. The system considers multiple levels of similarity, including word-level, semantic-level, and text-level features, to enhance accuracy. This approach reduces redundancy and improves resource utilization by grouping similar complaints. However, it primarily focuses on textual similarity and does not incorporate multi-modal data such as images and location.

## 2.5 Summary

This chapter reviewed various existing systems and research works related to civic grievance management. While traditional systems improved accessibility, recent AI-based approaches have introduced automation and intelligence into complaint handling. However, the lack of a unified and fully integrated solution highlights the need for an advanced system.

CivicMind builds upon these existing works by combining multiple technologies into a single platform, offering a comprehensive, efficient, and scalable solution for modern urban governance.

*Table 1.1: Literature Survey*

| Year | Author(s) | Title | Technology Used | Description |
|---|---|---|---|---|
| 2025 | S. Shastri, A. Bhide, S. Khan, T. Ashtankar, T. S. Dhillon, S. Awsarmol | AI-Powered Complaint Analyzer (AIPCA) Using Unified Framework Distil-BERT for Cross-Domain Grievance Management | Distil-BERT, Multi-Task Learning (MTL), Domain Adaptation (DA), Python/Flask, React, MongoDB, Knowledge Distillation | Web-based platform using fine-tuned Distil-BERT for automated complaint categorization and urgency detection. Achieves 92% accuracy with cross-domain functionality. Reduces MTTR from 72 hours to 12 hours through automated routing and priority assessment. |
| 2024 | A. Saha, A. Chaure, A. Bodhe, T. D. Bhagat | Automated Complaint Classification and Routing Using NLP and Machine Learning | TF-IDF, BERT Embeddings, Logistic Regression, SVM, Random Forest, KNN, Multinomial Naive Bayes, Python | System for automating classification and routing of public complaints using NLP and ML. Employs multiple classifiers (LR, SVM, RF, KNN, Naive Bayes, BERT) with TF-IDF and BERT embeddings. Handles multilingual data efficiently for departmental routing. |
| 2025 | C. Ashwitha, K. Vani | Smart Civic Complaint Analyzer Using Natural Language Processing | Python, Streamlit, TF-IDF, Multinomial Naive Bayes, Pandas, Excel/CSV storage | AI-enabled web interface for civic complaint analysis. Automatically categorizes complaints (Water, Roads, Electricity, Sanitation), detects urgency levels (High, Medium, Low), and extracts location. Features interactive dashboard with charts and word clouds for municipal officials. |
| 2019 | S. Kumar, S. Atreja, A. Singh, M. Jain | Adversarial Adaptation of Scene Graph Models for Understanding Civic Issues | Scene Graph Generation, Adversarial Training (ADDA), Faster R-CNN, MotifNet, LSTM, Deep Learning, Image Processing | Novel approach using adversarial training of scene graph models to generate Civic Issue Graphs from images. Uses Faster R-CNN for object detection and MotifNet with bidirectional LSTMs. Enables understanding of civic issues from images without labeled training data through cross-domain adaptation. |
| 2022 | P. D. C, V. B | Smart City Complaint Management System Using Android Application | Android Application, Web Portal, GPS Tracking, Image Upload, Mobile Technology | Android and web-based application enabling citizens to register civic complaints with images, location data, and text descriptions. |

| | | | | |
|---|---|---|---|---|
| | | | | Features GPS tracking for precise location identification, complaint status tracking, and direct communication channel between citizens and municipal departments. |
| 2025 | Kuruba Hemanth Kishore, Sirisha VL, M Naveen, Shashank Hosamani, Dr. Sapna B Kulkarni | CivicWatch: A Community-Driven Platform For Reporting And Resolving Local Infrastructure Issues | Firebase, Google Maps API, Zapier, ImgBB, Android app | Citizen-focused mobile app for reporting civic issues like damaged roads and streetlights with GPS, images, community validation, and admin dashboard for tracking and resolution. |
| 2021 | Sanket Nimase, Akshay Kanawade, Shital Kamble, Shubhangi Khemnar | Public Complaint Sorting for Smart City using Image Processing | CNN, Transfer Learning (ResNet, Inception), Python, Flask | Web application enabling citizens to report civic issues via images; uses ML image processing to automatically sort complaints to municipal departments, reducing manual sorting. |
| 2017 | Xiaolan Cui, Shuqin Cai, Yuchu Qin | Similarity-based approach for accurately retrieving similar cases to intelligently handle online complaints | Case-Based Reasoning (CBR), Ontology, WordNet, Semantic Similarity Measures | CBR framework using multi-level similarity (word, sense, text) measures on complaint products, problems, and content to retrieve reference solutions for new online complaints. |

# Chapter 3

# Limitations in Existing Systems

The field of civic grievance management systems has witnessed significant advancements over the years, driven by the increasing need for efficient communication between citizens and municipal authorities. Researchers have explored various approaches including web-based complaint systems, mobile applications, Natural Language Processing (NLP), and Computer Vision techniques to improve complaint handling processes. These systems provide valuable insights into methodologies for complaint submission, classification, routing, and resolution.

However, despite these advancements, most existing systems suffer from several limitations such as lack of automation, absence of multi-modal analysis, inefficient handling of duplicate complaints, and limited transparency. Many systems rely either on textual analysis or image-based processing, but fail to integrate both effectively. Additionally, issues such as manual intervention, lack of intelligent assignment, and poor scalability continue to affect system performance.

To better understand these limitations and identify research gaps, a detailed survey of relevant literature has been conducted. The selected research papers have been analyzed based on their contributions, methodologies, and shortcomings. The following table presents a structured comparison of existing systems, highlighting their key features and limitations.

*Table 3.1: Summary of limitations of existing systems*

| Title | Conference / Journal and Information Details | Key Points | Improvements Proposed | Citation |
|---|---|---|---|---|
| AI-Powered Complaint Analyzer Using Distil-BERT | IJRASET, 2025 | Uses Transformer (DistilBERT) for classification and urgency detection; improves routing efficiency | Focuses only on text data; lacks image validation and multi-modal integration | [1] |
| Automated Complaint Classification and Routing Using NLP | IJIRT, 2024 | NLP-based classification and routing using ML models like TF-IDF and BERT | Does not support image-based verification; limited fraud detection | [2] |
| Smart Civic Complaint Analyzer Using NLP | IJIRT, 2025 | Performs complaint classification, urgency detection, and dashboard analytics | Relies only on textual data; lacks real-time multi-modal validation | [3] |
| Adversarial Adaptation for Civic Issue Understanding | ACM Conference, 2019 | Advanced image understanding using scene graphs; multi-modal dataset | Complex implementation; not integrated with full complaint workflow | [4] |
| Smart City Complaint Management System | IJMRSET, 2022 | Web and Android-based system with complaint submission, image upload, and tracking | Manual classification and assignment; lacks AI automation | [5] |
| CivicWatch: Community-Driven Platform | IJCRT, 2025 | Geo-tagged complaint reporting, community validation, real-time tracking | No intelligent classification or automation; depends on manual validation | [6] |
| Public Complaint Sorting Using Image Processing | IJARCCE, 2021 | Uses CNN and image processing for issue detection and routing | Only image-based; ignores textual context; limited accuracy | [7] |
| Similarity-Based Complaint Retrieval Approach | Research Paper (Elsevier) | Uses case-based reasoning and text similarity for duplicate detection | Focuses only on text similarity; does not include image or location data | [8] |

# Chapter 4

# Problem Statement & Objectives

## 4.1 Problem Statement

Urban local bodies and municipal corporations are increasingly facing challenges in efficiently managing citizen grievances due to rapid urbanization and growing population density. The volume of complaints related to infrastructure, sanitation, water supply, electricity, and public safety has significantly increased, making traditional complaint management systems inadequate.

Existing grievance redressal systems are largely manual or semi-digital, relying heavily on human intervention for complaint categorization, verification, and assignment. This results in several inefficiencies such as delayed response times, duplicate complaints, lack of prioritization, and poor coordination among departments. Furthermore, these systems do not provide mechanisms for validating the authenticity of complaints, making them vulnerable to fraudulent or irrelevant submissions.

Another critical issue is the absence of intelligent prioritization. All complaints are treated equally regardless of urgency, which can lead to delays in addressing critical issues such as public safety hazards or infrastructure failures. Additionally, manual task assignment leads to uneven workload distribution among employees, causing inefficiencies in resource utilization. Transparency is also a major concern, as citizens often lack real-time updates regarding the status of their complaints. This reduces trust in governance systems and discourages civic participation.

Therefore, there is a need for an intelligent, automated, and centralized grievance management system that can:

- Accurately classify complaints
- Detect urgency and prioritize issues
- Validate complaint authenticity
- Eliminate duplicate submissions
- Automatically assign tasks based on workload
- Provide real-time tracking and transparency

The proposed system, CivicMind, aims to address these challenges by leveraging Artificial Intelligence techniques such as NLP and Computer Vision to create an efficient, scalable, and citizen-centric grievance redressal platform.

## 4.2 Objectives

The primary objective of the CivicMind system is to develop an AI-powered platform that enhances the efficiency and effectiveness of municipal grievance management through automation and intelligent decision-making. The specific objectives of the project are as follows:

- Automated Complaint Classification - To develop an NLP-based model that automatically classifies complaints into relevant sectors such as water, roads, sanitation, and electricity.
- Urgency Detection and Prioritization - To identify the severity of complaints and prioritize them based on urgency levels (low, medium, high).
- Image Validation and Fraud Detection - To implement a CNN-based model that verifies the authenticity and relevance of images submitted with complaints, reducing fraudulent or misleading entries.
- Intelligent Complaint Routing - To automatically route complaints to the appropriate municipal department based on classification results.
- Workload-Based Employee Assignment - To design an intelligent system that assigns complaints to employees based on their availability, expertise, and current workload.
- Duplicate Complaint Detection - To identify and merge duplicate complaints using location-based and semantic similarity techniques.

- Real-Time Tracking and Notifications - To provide citizens with real-time updates and status tracking for submitted complaints.
- Centralized Multi-City Platform - To develop a scalable system capable of supporting multiple municipal corporations within a unified platform.
- Improved Transparency and Accountability - To enhance governance by ensuring clear communication, visibility, and accountability in the complaint resolution process.

# Chapter 5

# Proposed System & Methodology

## 5.1 System Overview

The proposed system, CivicMind, is an AI-powered centralized grievance management platform designed to transform traditional municipal complaint handling into an intelligent, automated, and citizen-centric process. The system aims to bridge the gap between citizens and municipal authorities by providing a seamless digital interface supported by advanced Artificial Intelligence techniques.

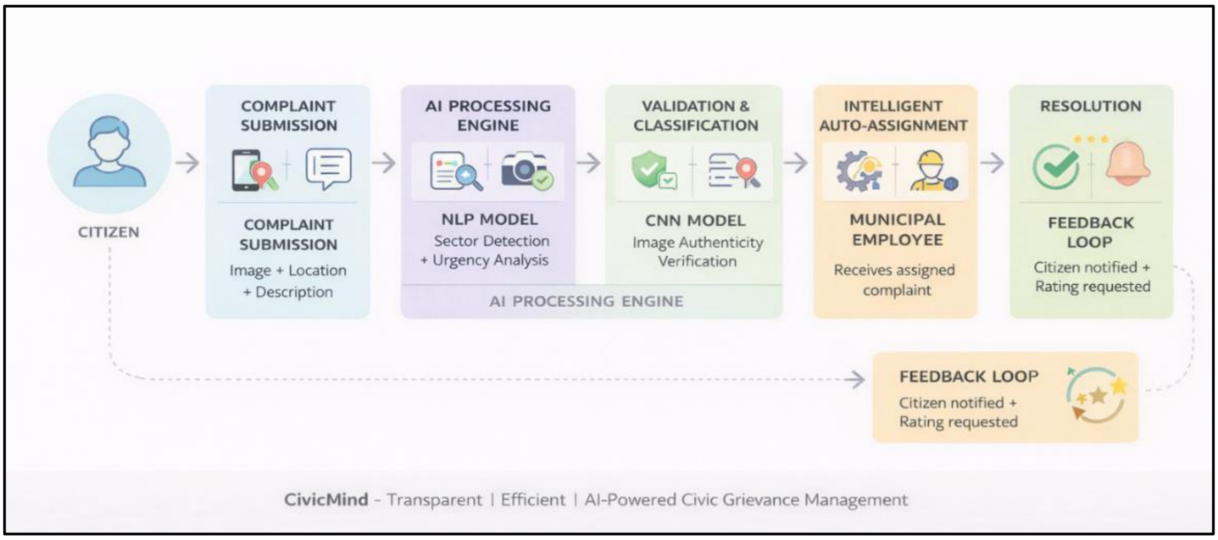


*Figure 1: Overview of the system flow*

Once the complaint is submitted, it is forwarded to the AI processing engine, which forms the core intelligence of the system. This engine consists of two major components: a Natural Language Processing (NLP) model and a Convolutional Neural Network (CNN) model. The NLP model analyzes the textual description to extract meaningful features such as complaint category (sector) and urgency level, while the CNN model processes the uploaded image to verify its authenticity and determine whether it corresponds to a valid civic issue.

Following this, the system performs validation and classification, where outputs from both models are combined. This stage ensures that the complaint is genuine and accurately categorized. Any inconsistency between textual and visual data can be flagged as a potential fraudulent or misleading complaint, improving the reliability of the system. After validation, the system executes intelligent auto-assignment, where the complaint is automatically routed to the appropriate municipal department. Additionally, the system assigns the complaint to a suitable municipal employee based on parameters such as workload, availability, and domain expertise. This eliminates manual intervention and ensures optimal resource utilization.

In the resolution phase, the assigned employee reviews the complaint, performs necessary actions, and updates the status within the system. The system continuously synchronizes updates across all interfaces. Finally, the system incorporates a feedback loop, where citizens are notified about the resolution and can provide feedback or ratings. This feedback is used to evaluate system performance and improve future service delivery.

Overall, the CivicMind workflow ensures automation, transparency, efficiency, and improved citizen engagement, making it highly suitable for modern smart city applications.

## 5.2 System Architecture

The CivicMind system is designed using a three-tier architecture, which ensures modularity, scalability, and efficient separation of responsibilities across different layers. This architectural approach allows independent development, deployment, and scaling of system components, making it suitable for real-world municipal applications. The architecture integrates a dedicated machine learning microservice to enable intelligent complaint classification and validation using both textual and visual data. The backend layer coordinates communication between the frontend, database, and AI services, ensuring smooth data flow and real-time processing. Additionally, the system incorporates geospatial analysis and deduplication mechanisms to improve accuracy and prevent redundant complaint handling. This layered design enhances system reliability, maintainability, and performance, making it suitable for large-scale civic applications.

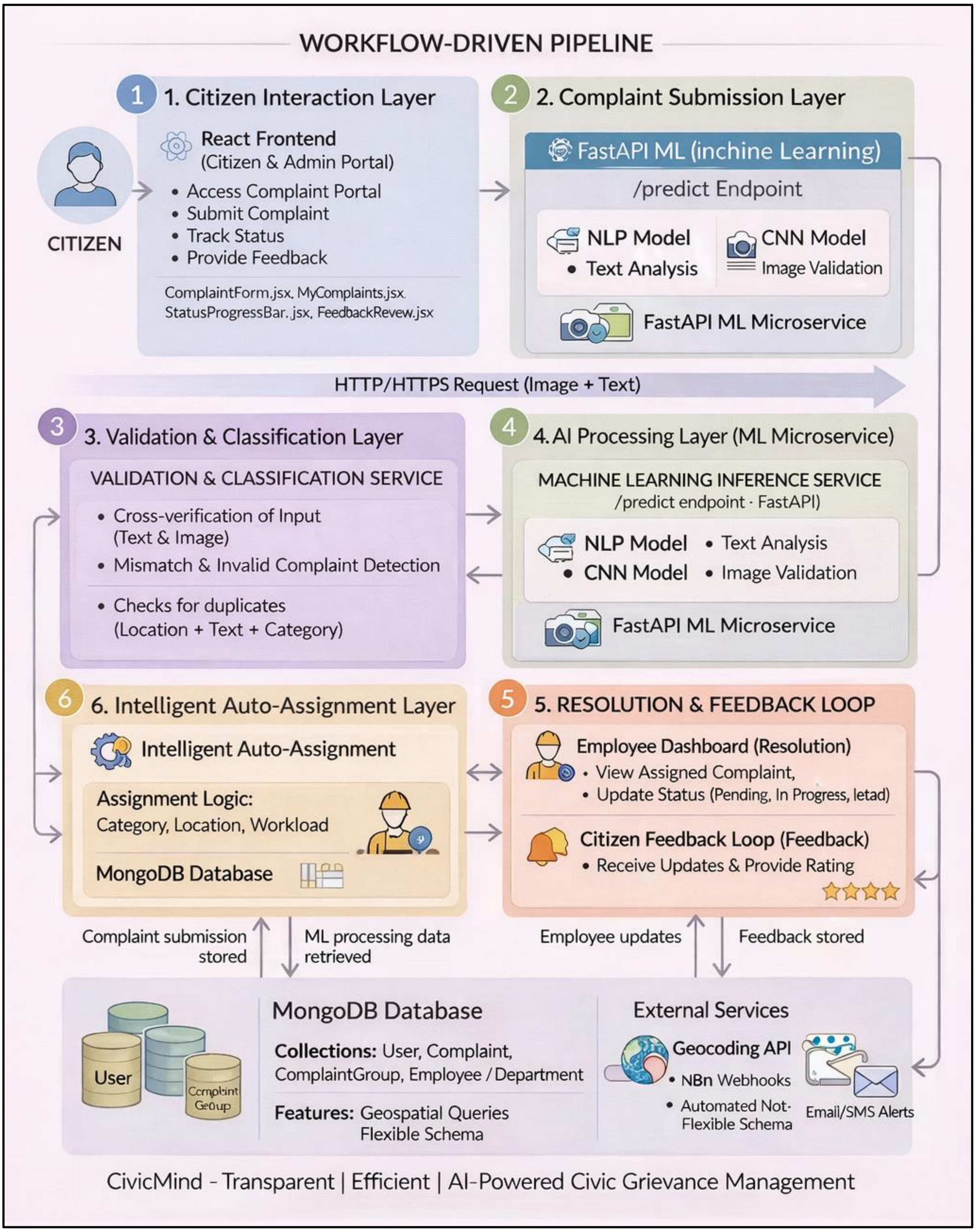


*Figure 2: Overview of the System Architecture*

### 5.2.1 Presentation Layer

The Presentation Layer acts as the user interaction interface and is responsible for delivering a seamless and intuitive user experience. It consists of two primary components:

- **Citizen Portal:**

This interface enables users to submit complaints, upload images, share location data, track complaint status in real time, and provide feedback after resolution. It also includes features such as notifications and access to official municipal announcements.

- **Municipal Dashboard:**

This interface is designed for administrative users and municipal employees. It provides tools for monitoring complaint queues, analysing performance metrics, managing departments, and viewing employee workloads. It enhances decision-making through analytics and visualization.

This layer is typically implemented using modern frontend technologies such as React.js and Tailwind CSS, ensuring responsiveness and usability across devices.

### 5.2.2 Application Layer

The Application Layer serves as the backbone of the system, handling all business logic and communication between the frontend and backend services. It is implemented using Node.js and Express.js, providing a robust and scalable backend environment. Key components include:

- **API Gateway:** Handles RESTful API requests from the frontend and routes them to appropriate services.
- **Authentication & Authorization Module:** Ensures secure user access using mechanisms such as JWT tokens, session management, and role-based access control.
- **Business Logic Module:** Processes the entire complaint lifecycle, including validation, classification requests, routing logic, notification handling, and status updates.
- **Real-Time Communication:** Supports notification services and real-time updates using Web Sockets or similar technologies.

This layer plays a critical role in orchestrating the workflow and ensuring smooth integration between different system components.

### 5.2.3 Data & Machine Learning Layer

This layer is responsible for data storage, retrieval, and AI-based processing. It combines database systems with machine learning inference services to enable intelligent decision-making.

- **Database (MongoDB):** Stores structured and unstructured data including user information, complaint records, employee details, department mappings, and analytics data. MongoDB's geospatial indexing supports efficient location-based queries.

- **File Storage (GridFS):** Handles storage of large files such as complaint images.
- **Machine Learning Inference Service (FastAPI):** This service performs real-time AI predictions and consists of:
  - NLP Model: Classifies complaint text into sectors and determines urgency levels.
  - CNN Model: Verifies image authenticity and detects civic issues.
  - Fraud Detection Module: Identifies mismatches between text and image inputs.
  - Intelligent Routing System: Maps complaints to departments and assigns employees based on workload.
- **External Integration (Maps API):** Provides geolocation services for accurate complaint tracking and mapping.

This layer ensures that the system remains intelligent, scalable, and capable of handling real-time data processing.

## 5.3 System Methodology

The methodology of the CivicMind system defines a comprehensive and structured framework for processing, analyzing, and resolving citizen complaints using Artificial Intelligence–driven automation. It represents a systematic workflow that integrates multiple technologies such as Natural Language Processing (NLP), Computer Vision (CNN), geospatial analysis, and intelligent decision-making mechanisms to ensure efficient grievance handling. Unlike traditional systems that rely heavily on manual intervention, the CivicMind methodology emphasizes automation, accuracy, and real-time responsiveness at every stage of the complaint lifecycle. The approach is designed to handle multi-modal data inputs, including text, images, and location, and to transform them into actionable insights through advanced machine learning models. By combining validation, classification, routing, and feedback mechanisms into a unified pipeline, the system ensures not only faster resolution of complaints but also improved transparency, accountability, and citizen satisfaction. The following steps describe the detailed methodology adopted in the CivicMind system.

### 5.3.1 Complaint Submission

The methodology begins with the complaint submission phase, where a citizen reports a civic issue through the system's user interface. The user provides a detailed textual description of the problem along with supporting image evidence and real-time location data captured using geolocation services. This multi-modal input ensures that the system receives sufficient contextual information to accurately interpret the nature and severity of the complaint.

The integration of image and location data enhances the credibility of the submission and enables precise identification of the issue's location.

### 5.3.2 Data Preprocessing

Once the complaint is received, the system performs preprocessing to prepare the input data for further analysis. The textual data is cleaned to remove noise, tokenized into meaningful units, and transformed into numerical representations suitable for NLP models. Simultaneously, the uploaded image undergoes preprocessing steps such as resizing, normalization, and enhancement to ensure compatibility with the CNN model. The location data is also validated and formatted to enable geospatial operations. This preprocessing stage ensures that all inputs are standardized and optimized for accurate model predictions.

### 5.3.3 NLP-Based Classification

In this stage, the processed textual data is analyzed using a Natural Language Processing model. The model extracts semantic meaning from the complaint description and classifies it into a relevant sector such as roads, sanitation, water supply, or electricity. Additionally, the model determines the urgency level of the complaint by analyzing keywords and contextual cues within the text. This classification enables the system to understand both the nature and priority of the issue, forming the basis for subsequent decision-making processes.

### 5.3.4 CNN-Based Image Validation

Parallel to text analysis, the system processes the uploaded image using a Convolutional Neural Network. The CNN model evaluates the authenticity and relevance of the image to determine whether it corresponds to a valid civic issue. It identifies visual patterns associated with common urban problems such as potholes, garbage accumulation, or water leakage. This step helps in filtering out irrelevant or misleading images, thereby improving the reliability and integrity of the complaint data..

### 5.3.5 Cross-Modal Validation (Fraud Detection)

After obtaining outputs from both the NLP and CNN models, the system performs cross-modal validation to ensure consistency between textual and visual inputs. The predicted category from the text is compared with the classification derived from the image. If both results are aligned, the complaint is considered valid and proceeds further in the workflow. However, if a mismatch is detected, the system flags the complaint as potentially fraudulent or misleading and may subject it to manual review. This mechanism significantly reduces false or spam complaints.

### 5.3.6 Intelligent Routing

Once the complaint is validated, it is routed to the appropriate municipal department based on the classification results. The system uses predefined mappings between complaint categories and municipal departments, along with geographical data, to ensure that the issue is directed to the correct authority. This automated routing eliminates delays caused by manual sorting and ensures that complaints reach the responsible department without unnecessary processing time.

### 5.3.7 Auto-Assignment

Following routing, the system performs intelligent assignment of the complaint to a municipal employee. The assignment process considers multiple factors such as employee availability, current workload, area of expertise, and geographical proximity. By distributing tasks efficiently, the system ensures balanced workload management and prevents overburdening of specific employees. This automated assignment enhances operational efficiency and speeds up the resolution process.

### 5.3.8 Resolution Process

In this phase, the assigned municipal employee reviews the complaint details and takes appropriate action to resolve the issue. The employee may conduct on-site verification, perform necessary repairs, or coordinate with other departments if required. Throughout this process, updates are recorded in the system to reflect the progress of the complaint. This ensures that all actions are documented and traceable.

### 5.3.9 Real-Time Tracking and Notifications

The system provides real-time updates to the citizen regarding the status of their complaint. Notifications are sent at various stages of the workflow, including assignment, progress updates, and resolution. The citizen can also track the complaint through the dashboard interface. This transparency improves user experience and builds trust in the system.

### 5.3.10 Feedback and Continuous Improvement

After the complaint is resolved, the system requests feedback from the citizen regarding the quality of service. The user can rate the resolution and provide comments. This feedback is stored and analyzed to evaluate system performance, identify areas for improvement, and enhance future service delivery. Over time, this feedback-driven approach contributes to continuous system optimization and better governance outcomes.

# Chapter 6

# Experimental Setup

## 6.1 Frontend Setup

The frontend of the proposed CivicMind system is developed using React.js to provide a responsive, interactive, and user-friendly interface for complaint submission, tracking, and management.

Navigation between different modules such as complaint submission, tracking, and dashboards is handled using client-side routing. The user interface is designed using Tailwind CSS to ensure responsiveness, consistency, and ease of use across different devices and screen sizes. The design focuses on simplicity and accessibility, allowing users to interact with the system efficiently.

### 6.1.1 Frontend Technologies Used

- **React.js –** Component-based user interface development
- **HTML5 –** Structure of web pages
- **CSS (Tailwind CSS) –** Styling and responsive design
- **JavaScript (ES6+) –** Client-side logic and interactivity
- **React Router DOM –** Navigation between pages
- **Geolocation API –** Capturing real-time user location
- **Axios / Fetch API –** API communication with backend

### 6.1.2 Frontend Structure

The frontend follows a modular architecture consisting of multiple components and pages:

- Complaint submission page for uploading images, entering description, and capturing location
- Dashboard page for tracking complaint status in real-time
- Admin panel for monitoring complaints, employee workload, and analytics
- Employee dashboard for viewing assigned tasks and updating status
- Notification and feedback modules for user interaction

This modular structure ensures efficient communication between user actions and backend services, enabling smooth user experience.

## 6.2 Backend Setup

The backend of the CivicMind system is implemented using Node.js and Express.js, which handle all core functionalities including API routing, authentication, business logic, and complaint lifecycle management. The backend is responsible for processing requests from the frontend, interacting with the database, and coordinating with the machine learning service. MongoDB is used as the primary database to store complaint records, user data, employee information, and system analytics. The backend also incorporates geospatial querying for location-based operations such as complaint deduplication and jurisdiction mapping.

In addition to the main backend, a separate Machine Learning backend is developed using Python and FastAPI. This microservice handles AI-based inference tasks such as complaint classification, urgency detection, and image verification using deep learning models.

### 6.2.1 Backend Technologies Used

- **Node.js –** Backend runtime environment
- **MongoDB –** NoSQL database for storing system data
- **JWT (JSON Web Tokens) –** Authentication and authorization
- **FastAPI (Python) –** Machine learning inference service
- **Deep Learning (LSTM/Transformer-based NLP model) –** NLP model implementation
- **PyTorch (ResNet-50) –** Image classification model
- **OpenStreetMap API –** Location and geospatial services

### 6.2.2 Backend Structure

The backend is organized into modular components where each module performs a specific function in the complaint lifecycle:

- **Authentication Module –** Handles user login, registration, and access control
- **Complaint Controller –** Manages complaint creation, updates, and retrieval
- **Deduplication Service –** Detects duplicate complaints using geospatial and text similarity
- **Assignment Service –** Assigns complaints to employees based on workload and availability
- **Notification Service –** Sends updates and alerts to users
- **ML Integration Module –** Communicates with FastAPI service for AI predictions

This modular structure improves scalability, maintainability, and system reliability.

### 6.2.3 API Communication

The system uses RESTful API communication to connect frontend requests with backend services and machine learning modules.

- The frontend sends complaint data (image + text + location) via POST requests
- FastAPI endpoint (/predict) processes NLP and CNN models
- Backend API (/api/complaints) handles complaint storage and processing
- JWT middleware ensures secure and authenticated requests
- Deduplication and assignment services are triggered during API execution
- Notifications are sent to dashboards using asynchronous updates

This API-driven architecture ensures smooth integration between all system components and enables real-time processing of complaints.

## 6.3 Development Environment

The development of the CivicMind system is carried out using a full-stack development environment that integrates frontend, backend, and machine learning components. Each module is developed and tested independently before integration to ensure system reliability. Local development is performed before deployment, allowing debugging, testing, and validation of all features such as complaint submission, AI processing, and API communication.

### 6.3.1 Software Requirements

- **Visual Studio Code –** Code editor
- **Windows Operating System –** Development platform
- **Node.js –** Frontend and backend runtime

- **Python 3.x –** Machine learning backend
- **MongoDB / MongoDB Atlas –** Database system
- **Postman –** API testing tool
- **Git –** Version control system
- **Web Browser (Chrome/Firefox) –** Testing and execution

### 6.3.3 Development Setup

The development setup includes configuration of both frontend and backend components before deployment.

- React application configured locally using Node.js environment
- Backend server set up using Express.js with REST APIs
- MongoDB database connected for data storage
- FastAPI service configured for ML inference
- Python dependencies installed for NLP and CNN models
- Image processing libraries configured for model input handling
- APIs tested using Postman before integration
- Local testing performed to verify UI navigation, API communication, and ML predictions

After successful local testing, all components are integrated to ensure seamless interaction between frontend, backend, and machine learning services.

# Chapter 7

# Results & Discussion

## 7.1 System Implementation Overview

The CivicMind system was successfully implemented as a full-stack web application integrating frontend, backend, and machine learning components. The system demonstrates seamless interaction between the React-based user interface, Node.js backend, and FastAPI-based machine learning service.The implementation supports real-time complaint submission, automated classification, intelligent routing, and live status tracking. All modules were integrated and tested to ensure smooth communication between components. The system effectively handles multi-modal inputs such as text, images, and location data, and processes them using AI models to generate accurate predictions. The deployment of role-based dashboards ensures that citizens, municipal authorities, and field employees can interact with the system according to their respective responsibilities.


*Figure 3: CivicMind System Interface Overview*

## 7.2 Functional Validation

The system was tested across different scenarios to validate its core functionalities. The complaint submission module was tested by uploading images along with textual descriptions and location data. The system successfully processed inputs and generated appropriate outputs through the AI pipeline. The NLP model correctly classified complaints into predefined sectors such as roads, sanitation, and water supply, while also identifying urgency levels. The CNN model effectively verified image authenticity and detected relevant civic issues. The integration between NLP and CNN models ensured accurate cross-validation and reduced the chances of fraudulent submissions. Additionally, the intelligent routing system successfully mapped complaints to appropriate departments, and the assignment module allocated tasks to employees based on workload and availability. The system also demonstrated efficient handling of duplicate complaints by grouping similar reports using location and text similarity analysis.

Real-time status tracking functionality was validated, allowing users to monitor complaint progress from submission to resolution. The feedback module successfully collected user responses, enabling performance evaluation of the resolution process. Overall, the system showed reliable performance, accuracy, and seamless integration of all modules under different test scenarios. The system also demonstrated efficient handling of duplicate complaints by grouping similar reports using location and text similarity analysis. Real-time status tracking functionality was validated, allowing users to monitor complaint progress from submission to resolution.

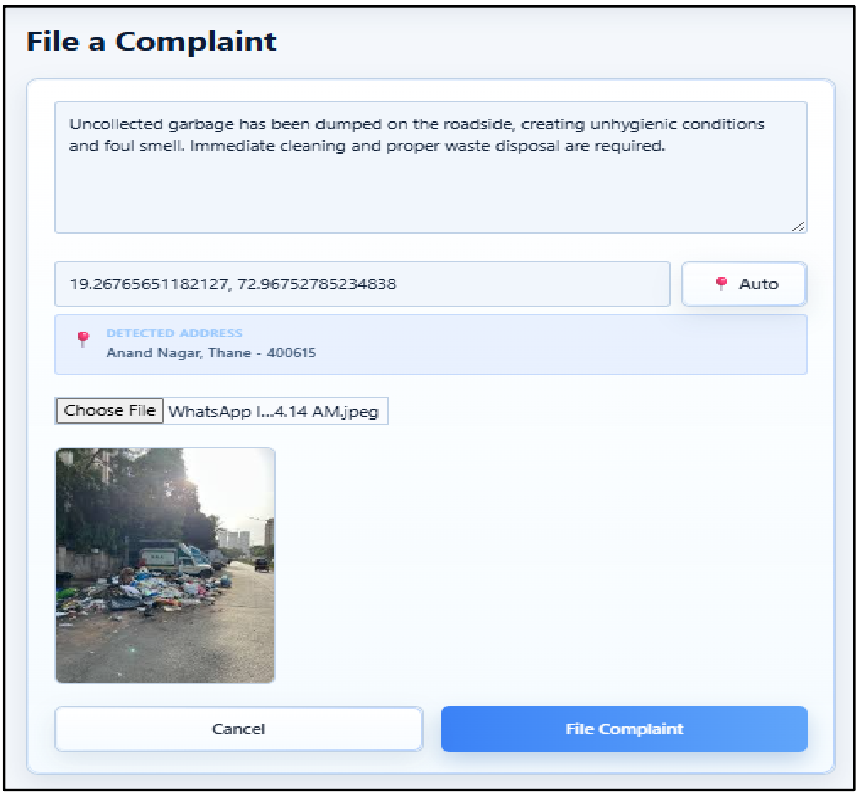


*Figure 4: Complaint Submission Page*

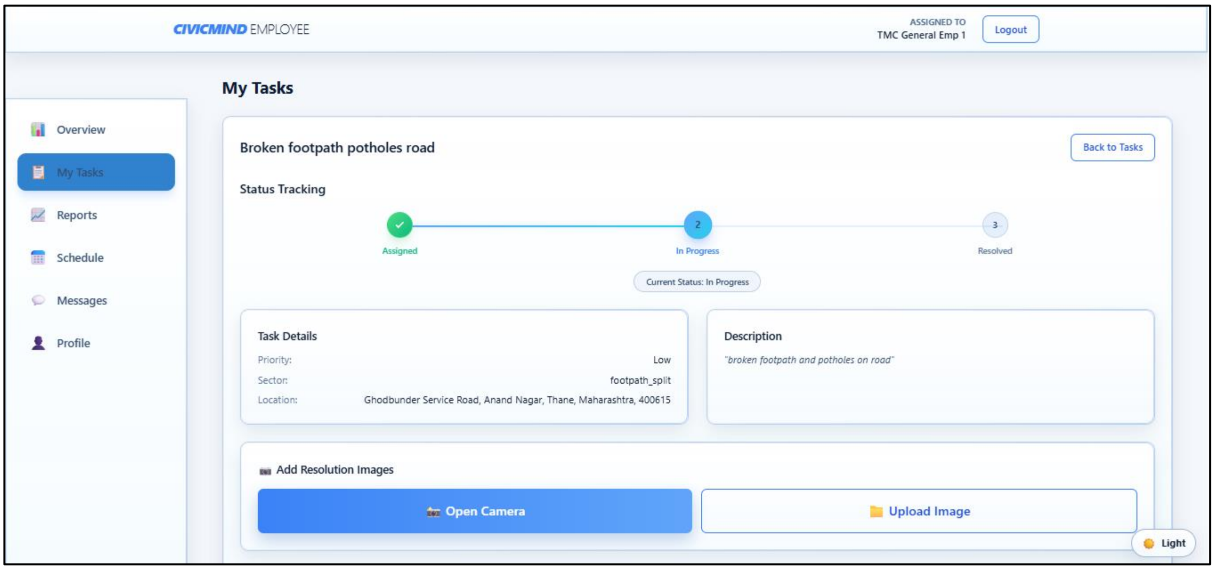


*Figure 5: Employee's Dashboard Of Ongoing Task*

## 7.3 Complaint Lifecycle Execution

To demonstrate the working of the system, a case study was conducted where a user submitted a complaint regarding a pothole on a road. The user uploaded an image, entered a description, and shared location data. The system processed the complaint using the NLP model, which classified it under the "Roads and Transportation" sector with a medium urgency level. The CNN model verified the image and confirmed that it matched a road-related issue. The system then checked for duplicate complaints in the same geographical area and grouped similar complaints if found. Following validation, the complaint was automatically routed to the relevant municipal department and assigned to an available employee.

The employee reviewed the complaint, updated the status after inspection, and resolved the issue. The citizen received real-time notifications and later provided feedback on the resolution. This case study demonstrates the effectiveness of the system in handling end-to-end complaint processing.

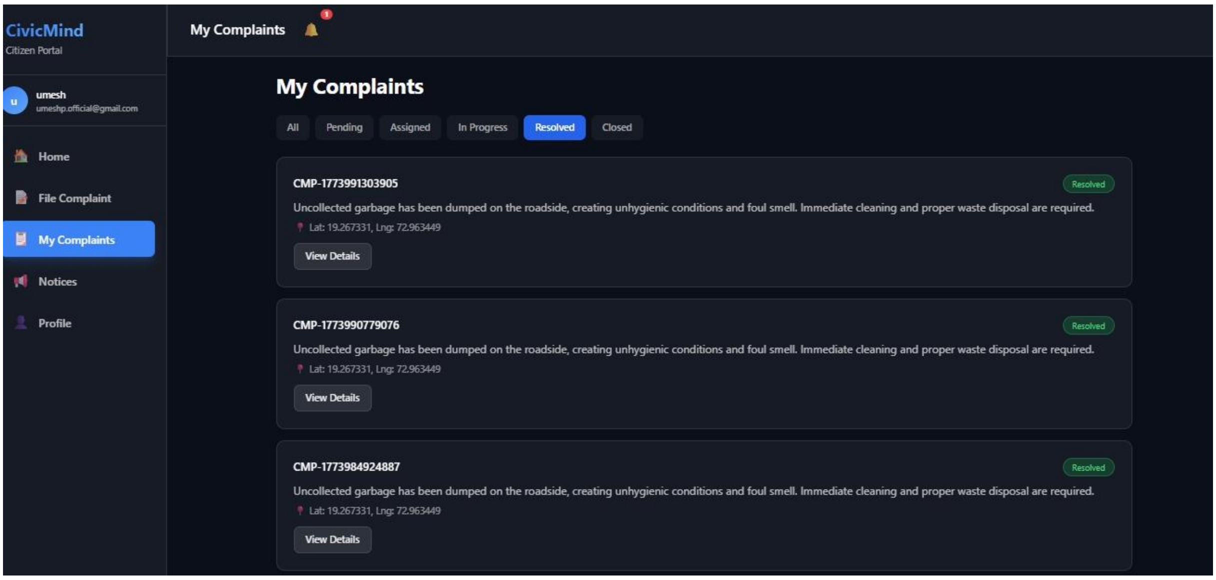


*Figure 6: Citizen's Dashboard of his complaints*

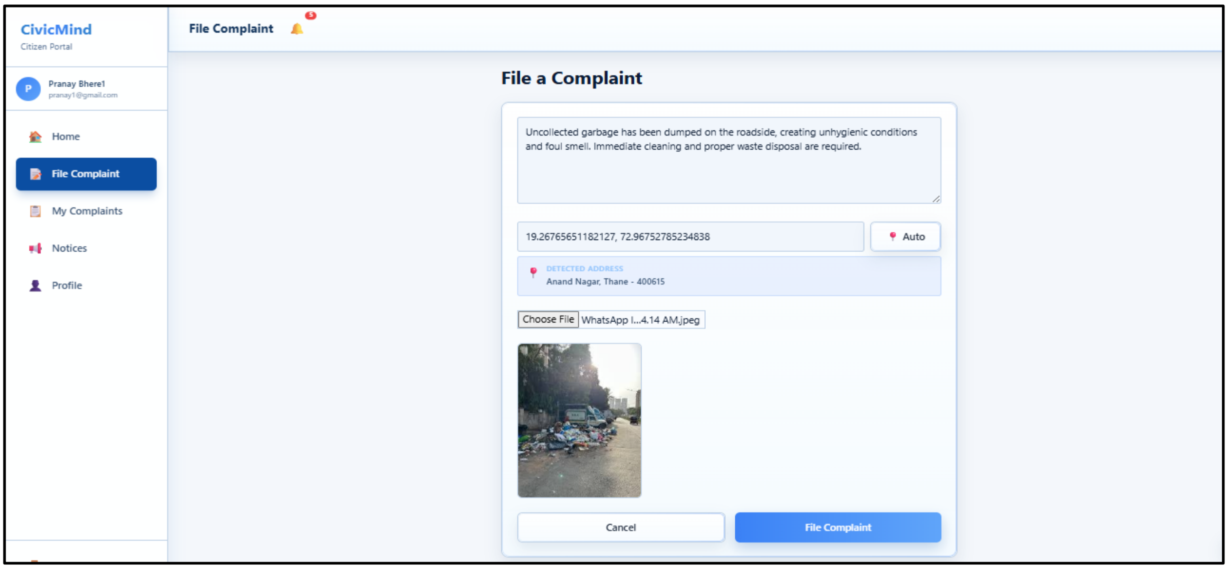


*Figure 7: Interface for issuing New Complaint*

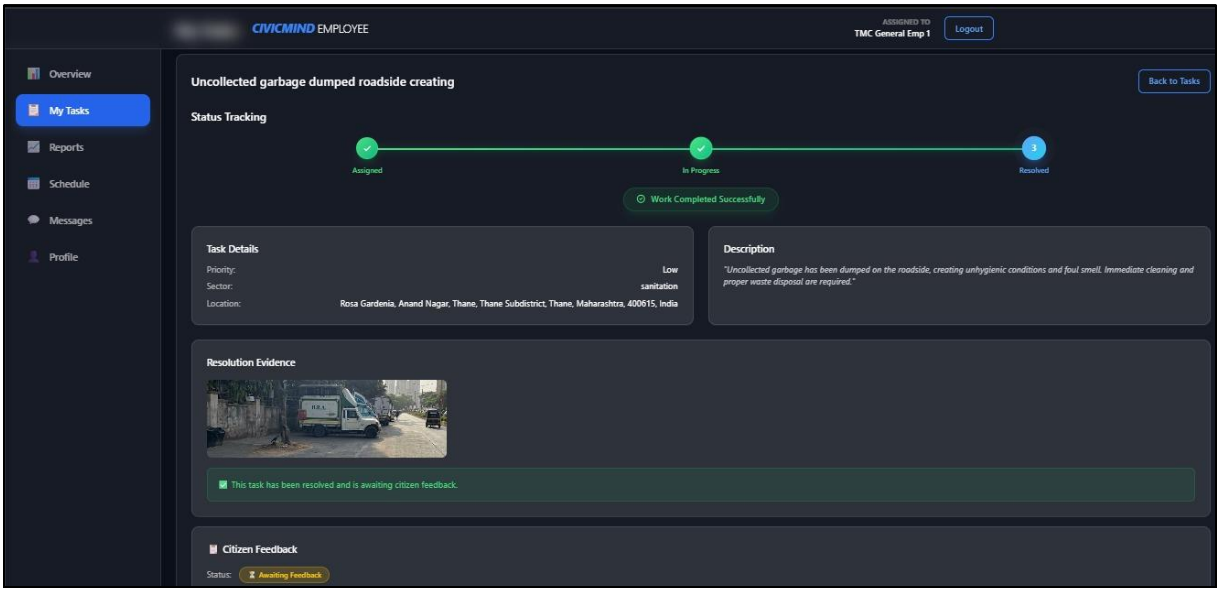


*Figure 8: Employee's View for tracking complaints progress*

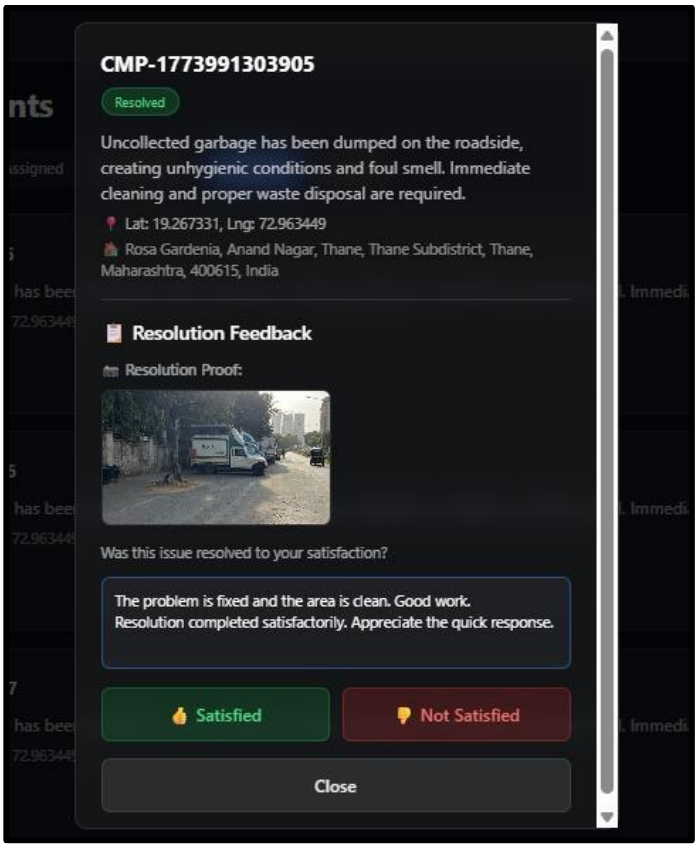


*Figure 9: Citizen's View after Complaint Resolution*

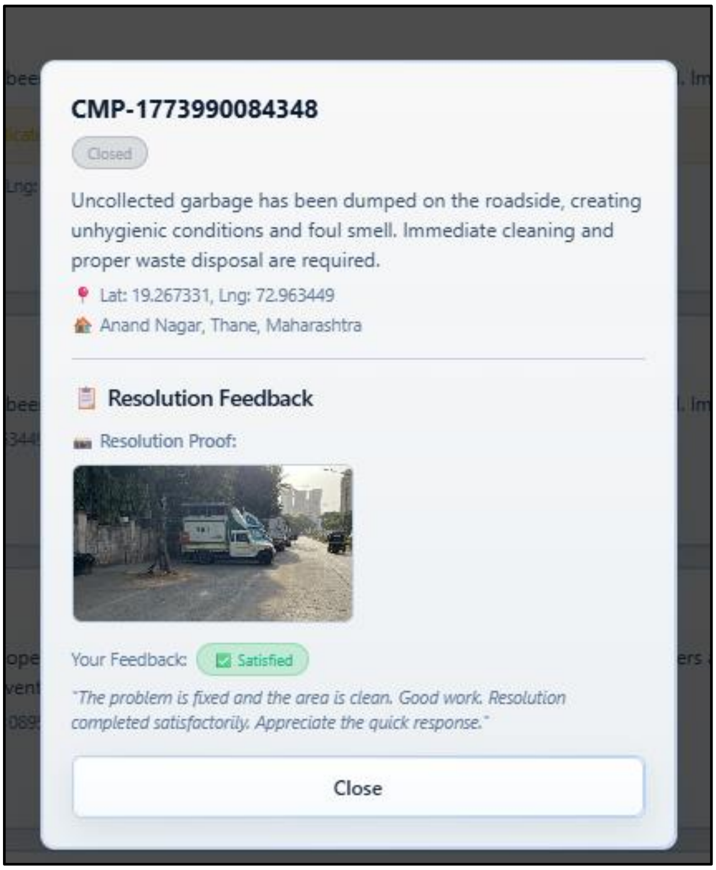


*Figure 10:Citizen's View after submitting feedback for resolution*

## 7.4 System Performance and Usability Analysis

The performance of the CivicMind system was evaluated based on multiple parameters including accuracy, response time, operational efficiency, and user experience. The integration of Artificial Intelligence models with a responsive web interface ensures both high system performance and ease of use.

From a technical perspective, the NLP model achieved an accuracy of approximately 92% in sector classification and around 89% in urgency prediction. Similarly, the CNN model demonstrated an accuracy of approximately 89% in image classification and validation tasks. The average inference time for processing a complaint was observed to be between 2 to 5 seconds, indicating efficient real-time processing capabilities. The deduplication mechanism effectively reduced redundant complaints by identifying similar issues using geospatial proximity and textual similarity. Additionally, the intelligent assignment system ensured balanced workload distribution among employees, thereby improving operational efficiency and reducing response delays. In terms of usability, the system provides a user-friendly and intuitive interface that simplifies complaint submission and tracking.

Citizens can easily report issues without requiring technical expertise, while municipal administrators and employees benefit from structured dashboards and automated workflows. Real-time notifications and status tracking enhance transparency by keeping users informed throughout the complaint lifecycle. The system also supports smooth operation under multiple concurrent requests due to its modular architecture and efficient backend processing.

Overall, the CivicMind platform successfully balances technical performance with user experience, ensuring efficient grievance handling while maintaining high levels of accessibility, transparency, and reliability.

## 7.4 Limitations

Despite its effectiveness, the CivicMind system has certain limitations. The accuracy of the NLP and CNN models depends on the quality and size of training data. In some cases, ambiguous text or unclear images may affect prediction accuracy. The system also relies on internet connectivity for real-time processing and communication. Additionally, extreme workload scenarios may require further optimization for large-scale deployment. Future improvements can focus on enhancing model accuracy, incorporating multilingual support, and improving scalability for handling large volumes of complaints.

# Chapter 8

# Conclusion & Future Scope

## 8.1 Conclusion

The CivicMind system successfully demonstrates the application of Artificial Intelligence in transforming traditional municipal grievance redressal mechanisms into an intelligent, automated, and citizen-centric platform. The project addresses critical challenges faced by existing systems, such as manual processing, lack of transparency, delayed response times, and inefficient resource allocation. By integrating advanced technologies such as Natural Language Processing and Computer Vision, the system is capable of automatically classifying complaints, detecting urgency levels, and verifying image authenticity. The implementation of cross-modal validation enhances the reliability of the system by reducing fraudulent or irrelevant submissions. Furthermore, the incorporation of geospatial-based deduplication ensures that redundant complaints are minimized, improving overall efficiency.

The intelligent routing and workload-based assignment mechanisms significantly enhance operational performance by ensuring that complaints are directed to the appropriate departments and assigned to available personnel in a balanced manner. The system also promotes transparency and accountability through real-time tracking, notifications, and feedback mechanisms, thereby improving citizen engagement and trust in governance.

Overall, CivicMind provides a scalable and efficient solution for modern urban governance, aligning with the goals of smart city initiatives. The system not only improves grievance handling but also contributes to better decision-making and resource management within municipal bodies.

## 8.2 Future Scope

Although the CivicMind system achieves its primary objectives, there are several opportunities for further enhancement and expansion to improve its capabilities and applicability. One potential area of improvement is the enhancement of machine learning models through the use of larger and more diverse datasets. This would improve the accuracy of complaint classification, urgency detection, and image validation. Additionally, incorporating multilingual Natural Language Processing capabilities would enable the system to support users from diverse linguistic backgrounds, making it more inclusive. The system can also be extended to include predictive analytics, where historical complaint data is analyzed to identify patterns and forecast potential civic issues. This would enable municipal authorities to take proactive measures rather than reactive responses.

Another area of future development is the integration of mobile applications to improve accessibility and provide features such as real-time notifications, offline complaint submission, and camera-based instant reporting. Integration with Internet of Things (IoT) devices, such as smart sensors for waste management or water leakage detection, can further enhance the system’s effectiveness. Scalability can be improved by deploying the system on cloud infrastructure with support for distributed processing and microservices architecture. This would enable the platform to handle large volumes of complaints across multiple cities simultaneously.

Finally, the system can be enhanced by incorporating advanced analytics dashboards, performance monitoring tools, and decision-support systems for administrators. These features would help municipal authorities make data-driven decisions and improve overall governance efficiency.

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